ENGR 211: Engineering Mechanics I - Statics (Fall 2024)   Instructor: Dr. [redacted] Ph.D.

[redacted]

Last updated: 8/26/2024   Tel: (use email)

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Textbook & Materials:

- R.C. Hibbeler, Engineering Mechanics: Statics, 15th Edition, Pearson, (ISBN-13: 9780134867298) (Older versions dating back to 12th edition are acceptable)
- Access to MATLAB. MATLAB is available on department computers, or you can download and install on your PC after making an account with mathworks.com using wheaton.edu or my.wheaton.edu email address for username.
- Access to Solidworks. Solidworks is available in the engineering lab and can be accessed online. Contact [redacted]

Course Description and Prerequisites:

Welcome to Statics! This course is truly the foundation of your undergraduate engineering training. So, take good notes, ask a lot of questions, and learn from your peers to master the ENGR 211 course materials – proper mastery of these topics will help prepare you for a successful transition into your upper-level mechanics courses and your future career as an engineer.

Systems of units; gravitation; Newton's laws of motion; equilibrium and free-body diagrams; particles, forces and moments; structures in equilibrium; centroids and center of mass; moments of inertia; friction; beam loadings; cables; fluids; virtual work and potential energy. Lecture and Laboratory. Prerequisite: PHYS 231. Pre or Corequisite: ENGR 334.

Student Outcomes:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.

Course Objectives:

After satisfactory participation in this course, students should expect to be able to identify with the following course-level learning outcomes, as well as concept mastery objectives and skill development objectives.

Course-Level Learning Outcomes:

1. I will be able to model, define, and analyze rigid-body, equilibrium systems using engineering, mathematical and physical principles
2. I will be able to design and create a mechanical system for a given problem that meets defined structural requirements

3. I will be able to apply multiple techniques to solve engineering mechanics problems (analytical, computational, empirical)
4. I will be able to autonomously learn mechanics concepts, apply them, and self-evaluate my understanding
5. I will be able to apply engineering mechanics principles to problems in various engineering disciplines
6. I will grow in my understanding of how engineering mechanics relates to serving Christ and the world.
7. I will better understand my own cultural identity, values, motivations, and biases and how this dictates my position in relationship to a cross-cultural team.

Concept Mastery Objectives:

1. I will be able to perform numerical calculations and perform conversion for different unit systems
2. I will be able to perform vector operations in cartesian and polar coordinates in 2D and 3D (dot product, cross product)
3. I will be able to generate a free body diagram for a given system in 2D and 3D
4. I will be able to explain the conditions for equilibrium and the type of equilibrium a system is in (stable, unstable, neutral)
5. I will be able to describe position, forces, and moments in vector forms in 2D and 3D
6. I will be able to determine the resultant of a force system including distributed forces and define an equivalent system
7. I will be able to explain the static determinacy of a system
8. I will be able to calculate the internal and reaction forces and moments in trusses, frames, and machines
9. I will be able to analyze systems which include frictional forces
10. I will be able to calculate the center of gravity and centroid (areal and mass) for systems of discrete particles or arbitrary shape and explain its relevance to engineering problems
11. I will be able to calculate the moment of inertia for systems (including composites)
12. I will be able to apply and understand the conditions for the principle of virtual work
13. I will be able to explain basic mechanical properties of structural materials (stress-strain, yield stress, ultimate stress)
14. I will be able to predict at a basic level, when a system will fail

Skill Development Objectives:

1. I will be able to use modern engineering tools to solve mechanics problems
2. I will be able to identify, explore, and explain the idealizations, approximations, and assumptions used in statics problems and their implications
3. I will be able to apply the design process to a stated problem (define, ideate, analyze, optimize, build, evaluate, iterate)
4. I will be able to optimize a structural design to meet given requirements using computational tools
5. I will be able to more productively collaborate within a diverse team to identify, research, and solve a problem while addressing and responding to different conflict, leadership, and communication styles, in a way that both promotes effectiveness and facilitates inclusion.

Grading:

This course emphasizes collaboration within diverse teams, learner engagement according to different cultural definitions, and personal ownership of the learning process. Classwork, homework, projects, and assessments are therefore meant to encourage engagement, positive collaboration, and enhance student learning of course topics.

Homework: 20%

- Homework assignments will be given online at Pearson Maylab and Mastering through LMS Canvas.
- Homework will be assigned for each chapter and please visit Pearson website often and complete timely.
- *Late Assignment Policy:* Late work is not accepted unless excused by instructor prior to the due date. Reduced credit may apply.
- *Academic Honesty:*
  Collaboration with your classmates on homework assignments is highly encouraged, however, each student must complete the answer to the question independently. Students are *not* permitted to copy or follow solutions manuals answers or other external sources in any way unless indicated.

Quizzes: 10%

- There are in-class and/or online quizzes that will be assigned approximately weekly.

Unit Exams: 30%

- There are 3-unit exam assessments.
- Unit exam topics and tentative dates are indicated on the course schedule.

Semester Project: 20%

- Projects will often be team-based and requires utilizing concepts learned from the course to solve an engineering problem. The project will emphasize practice with modeling, structural analysis, solid modeling, and 3D printing.
- Details and requirements for class projects will be communicated during class.

Final Exam: 15%

- The cumulative final exam will take place at the end of the semester and will cover the entire course material.

Participation/Engagement: 5%

- Participating in class discussions in a community-centered manner requires being on time, being attentive, contributing questions and comments, and having respect for others. It is own backgrounds, strive for regular attendance and collaborative performance in class.

Grading Scale:

Homework	*20%*
Unit Exams	*30%*
Quizzes	*10%*
Semester Project	*20%*
Final Exam	*15%*
Participation/Engagement	*5%*
Total	*100%*

Lab Safety & Engineering Support:

Engineering Makerspace safety training is required. See your section on the WheatonX Canvas instance. All students must train on general laboratory and tool safety guidelines prior to usage of any equipment. Our Engineering Lab Manager and Engineering Support TAs are available to help support project activities and provide equipment training and supervision. Their office hours will be posted outside the Wheaton Engineering Makerspace. Workshops will be provided which offer support and training for equipment and tools including laser cutter and 3D printing. Completion of assigned safety training and workshops are reflected in overall student project grades.

Attendance & Classroom Policies:

As this is a collaborative, project-based course, it is important that you are present and fully engaged during class periods. Each absence without valid excuse and prior permission from the instructor will result in a 1% final grade decrease. Leaving early before the class session is over, or being excessively late without permission will also count as an absence. Part of class time will be regularly devoted to project work. Always bring your project related materials. You may want to bring your own laptops to class for project work sessions. However, laptops must only be used for project related work.

Class Communication:

Course assignments, announcements and documents will be communicated in class or through Schoology. Questions or comments regarding assignments, projects, or class materials are welcome through the Schoology site or by e-mail. Grades will be posted through Schoology.

Use of Artificial Intelligence:

AI Writing tools are not permitted for any stage or phase of work in this class. If you use these tools, your actions would be considered academically dishonest and a violation of the Wheaton College Policy on Academic Honesty.

Accommodations

Wheaton College believes that disability is an indispensable part of the diversity of God's Kingdom. We work to provide equal access to college programs and activities as well as spaces of belonging for students with disabilities. Students are encouraged to discuss with their professors if they foresee any disability-related barriers in a course. Students who need accommodations in order to fully access this course's content or any part of the learning experience should connect with Learning and Accessibility Services (LAS) as soon as possible to request [redacted] accommodations process is dynamic, interactive, and completely free and confidential. Do not hesitate to reach out or ask any questions.

Diversity & Respect:

For academic discourse, spoken and written, the faculty expects students to use gender inclusive language for human being.

Confidentiality and Mandatory Reporting:

As an instructor, one of my responsibilities is to help create a safe learning environment on our campus. I also have a mandatory reporting responsibility related to my role as a faculty member. I am required to share information regarding sexual misconduct about a crime that may have occurred on Wheaton College's campus with the College. Confidential resources available to students include Confidential Advisors, the Counseling Center, Student Health Services, and the Chaplain's Office. More information on these resources and College Policies is available at www.wheaton.edu/sexualassaultresponse.

Course Schedule:

The following schedule is **tentative** (consider it a working document) and is subject to change at the discretion of the course instructor. Any major updates will be discussed in class and revised syllabus versions and assignments will be posted on Canvas modules.

			ENGR-211 Engineering Mechanics - Statics	11:35 AM – 12:45 PM (Meyer-143)
Week	**Date**	**Reading**	**Topic**	**HW/Lab Assignment & Activity**
1		**Ch 1-2**	**Introduction, Module 1 -Vectors Forces**	*HW will be given online thru Pearson Mastering. Pl visit the Pearson web often and complete timely.*
	Aug 28 (W)	1.1-1.2	Course Intro/Syllabus, Lab Introduction/General Safety (Dr. [redacted]	*Lab Intro. Plus General Safety*
	Aug 30 (F)	1.3-1.6	Units, Sig Figs, Vectors, Forces, Force Addition	
2	Sep 2 (M)		**Labor Day**	
	Sep 4 (W)	2.1-2.4	Vectors, Forces, Force Addition	*General Safety Quiz (Wheaton X) Due, Signup Glow forge 2 Due*
	Sep 6 (F)	2.5-2.9	Cartesian Vectors, Position Vectors, Dot Product	
3		**Ch 3-4**	**Module 2 - Equilibrium, Force & Moment Systems**	
	Sep 9 (M)	3.1-3.2	Particle Equilibrium, Free Body Diagrams	
	Sep 11 (W)	3.3	2D Equilibrium Systems Part 1	
	Sep 13 (F)		2D Equilibrium Systems Part 2	***Quiz #1***
4	Sep 16 (M)	3.4	3D Equilibrium Systems	
	Sep 18 (W)		Review and Equilibrium Problems	
	Sep 20 (F)		***Unit Exam 1 (Ch 1-3)***	***Unit Exam 1***
5	Sep 23 (M)	4.1-4.4	Moments, Cross Product	
	Sep 25 (W)	4.5-4.7	Moment of an Axis, Couples, Equivalent Systems	
	Sep 27 (F)	4.9	Distributed Loads	
6	Sep 30 (M)		Module 2 Problems & Review	***Quiz #2***
		Ch 5-6	**Module 3 - Rigid Body Equilibrium**	
	Oct 2 (W)	5.1-5.3	2D FBDs in Rigid Body Equilibrium, Equilibrium Equations	
	Oct 4 (F)	5.4-5.5	2 and 3 Force Members, Support Reactions, Internal Forces, Weight	
7	Oct 7 (M)	5.6-5.7	Constraints, Redundancy, Static Determinacy, Improper Constraints	
	Oct 9 (W)		Module 3A Review, ***3D-print workshop 1*** [redacted]	***3D-Print workshop***
	Oct 11 (F)		***3D print Workshop 2 by*** [redacted]	*Signup 3D Print 2 Due,*
8	Oct 14 (M)		***Unit Exam 2 (Ch 4-5)***	***Unit Exam 2***
	Oct 16 (W)	6.1-6.2	Trusses: Method of Joints	
	Oct 18 (F)	6.3	Trusses: Zero Force Members	
9	Oct 21 (M)		**Fall Break**	
	Oct 23 (W)			

	Oct 25 (F)	6.4	Method of Sections	
10	Oct 28 (M)	6.6	Frame & Machine	
	Oct 30 (W)		***Makerspace:*** *Balsawood Bridge Project Kickoff by* [redacted]	***Balsawood Bridge Project Kickoff***
	Nov 1 (F)		Chapter Review & ***Quiz #3***	***1) Bridge Design & BOM (10/28-11/8)***
11		**Ch 7-8**	**Module 4 – Internal Forces, Friction**	
	Nov 4 (M)	7.1	Internal Loadings	
	Nov 6 (W)	7.2	Shear & Moment Diagrams	
	Nov 8 (F)	7.3	Relations Among Distributed Load, Shear, & Moment	
12	Nov 11 (M)		***Makerspace:*** *Balsawood Bridge Design by* [redacted]	***2) Drawing & BOM due (11/11)***
	Nov 13 (W)	8.1-8.2	Dry Friction	***3) Bridge Construction (11/11-22)***
	Nov 15 (F)	8.3-8.5	Wedges & Flat Belts	
13	Nov 18 (M)		*Module 4 Review &* ***Quiz #4, Miter Saw Workshop*** *(10 min.)*	*Miter Saw Workshop by* [redacted]
	Nov 20 (W)	9.1	Center of Gravity, Center of Mass, Centroid	(**Guest Lecturer**)
	Nov 22 (F)		***Unit Exam 3 (Ch 6-7-8)***	***Bridge Impound by 4:00 PM***
		Ch 9-11	**Module 5 - CG, Centroids, Moment of Inertia, Virtual Work**	
14	Nov 25 (M)	9.2	Composite Bodies	*Signup Miter Saw Due*
	Nov 27 (W)		**Thanksgiving Break**	
	Nov 29 (F)			
15	Dec 2 (M)		***Makerspace: Bridge competition Tests*** [redacted]	***4) Bridge Compete Testing (12/2)***
	Dec 4 (W)	10.1-10.4	Area Moments of Inertia	(**Guest Lecturer**)
	Dec 6 (F)	10.8	Mass Moment of Inertia, ***Quiz #5***	
16	Dec 9 (M)	11.1-11.3	Principle of Virtual Work	
	Dec 11 (W)	11.4-11.7	Conservative Forces/Potential Energy Method	
	Dec 13 (F)		Module 5 Problem & Review, and Final Review	
17	TBD		***Project Presentations*** [redacted]	*Project Report Due*
	TBD		**Final Exam on Modules 1-5**	*Final Exam*